RIGIDITY OF SYMMETRIC HOLOMORPHIC FUNCTIONS ON INFINITE-DIMENSIONAL SEQUENCE SPACES

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Abstract. We study holomorphic functions on permutation-invariant subsets of an infinite-dimensional sequence space that are invariant under all finite permutations of coordinates. Working on domains modeled on the space c_0 of null sequences, we show that permutation symmetry forces the first differential to vanish along every constant tail: the coordinate derivatives agree there and an averaging argument annihilates their common value. This yields rigidity—constancy—on domains of constant sequences, and more generally on any c_{00}-chain-connected domain on which the differential vanishes on the finitely supported directions. The method has a sharp boundary: a precise obstruction prevents the elementary argument from reaching the finitely many exceptional coordinates of a non-constant point, so unconditional rigidity on general eventually-constant domains remains open. In the other direction, an explicit example—any Banach limit on ℓ^∞—shows that the chain-connectedness and boundedness hypotheses cannot be dropped: mere topological connectedness admits non-constant symmetric entire functions. We record the vector-valued corollary and discuss the motivating application to holomorphic cross-sections of the zero-set map, isolating the lifting problem that remains open.

1. Introduction

In finite dimensions the holomorphic functions on $\mathbb{C}^n$ invariant under all permutations of coordinates form a rich algebra, freely generated by the elementary symmetric polynomials $e_1, \ldots, e_n$. In infinite dimensions this richness is sharply curtailed: on the natural sequence spaces built from null sequences, a symmetric holomorphic function is severely constrained, and on the basic domains is forced to be constant. The mechanism is that the elementary symmetric functions $e_k = \sum_{|I|=k} \prod_{i \in I} x_i$ cease to converge for generic sequences, and this failure is visible already at the level of first derivatives.

This rigidity phenomenon holds under precise hypotheses, and it has a definite boundary beyond which it fails. That boundary is real rather than apparent: the global statement that *every* symmetric holomorphic function on a connected permutation-invariant subset of ℓ^∞ is constant is false. Any Banach limit furnishes a counterexample (Example 5.2). The rigidity is a feature of the null-sequence geometry, not of symmetry alone.

The main result, informally. Write c_{00} for the finitely supported sequences and c_0 for the null sequences, so that $\overline{c_{00}} = c_0$ in the supremum norm. Theorem 4.1 states: if X is a locally box-open, permutation-invariant, c_{00}-chain-connected subset of a translate of c_0, and $f : X \to \mathbb{C}$ is admissibly holomorphic with bounded differential, invariant under all finite permutations, *and* has differential vanishing on c_{00} at every point, then f is constant. The last hypothesis is essential. The analytic core discharges it unconditionally at constant points—hence yields outright rigidity on domains of constant sequences—and along the constant tail

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of any eventually-constant point, but *not* at the finitely many exceptional coordinates of a non-constant point, where a definite obstruction intervenes (Remark 4.3).

The proof of the vanishing has two elementary movements. First, at any point x whose coordinates are eventually equal to a constant, the coordinate derivatives $Df(x)[e_i]$ agree for all indices i in the constant tail, because the transposition of two tail indices fixes x. Second, averaging N such tail directions produces a perturbation of supremum norm $1/N \to 0$; a bounded differential then forces the common tail value to vanish. Constancy then propagates from vanishing derivatives along finitely supported segments, and c_{00}-chain-connectedness spreads it across X.

Two structural subtleties. Two features of the setting govern both the reach and the limits of the argument.

The base point moves under permutation. With the coordinate action $(x \circ \sigma)_n = x_{\sigma(n)}$, the transposition τ_{ij} satisfies $(x + \varepsilon e_i) \circ \tau_{ij} = (x \circ \tau_{ij}) + \varepsilon e_j$, which equals $x + \varepsilon e_j$ only when $x_i = x_j$. In general one obtains

$$Df(x)[e_i] = Df(x \circ \tau_{ij})[e_j],$$

an identity relating the differentials at *two different* base points. The conclusion "all coordinate derivatives at x coincide" is therefore valid at points fixed by the relevant transpositions—in particular along the constant tail of an eventually-constant sequence—but not, by this step alone, at an arbitrary point.

Finitely supported segments do not connect ℓ^∞. A finite chain of c_{00}-perturbations alters only finitely many coordinates, so it can never join 0 to $(1, 1, 1, \ldots)$. Vanishing of the differential on c_{00} controls f only within a single coset of $\overline{c_{00}} = c_0$; across cosets, symmetric holomorphic functions may vary freely, and Example 5.2 shows the global statement is false on ℓ^∞.

Relation to existing literature. The principle that infinite symmetry suppresses analytic variation is familiar in spirit; it is reminiscent of de Finetti-type rigidity in probability, where exchangeability forces a mixture representation. The role of the null-sequence geometry, and the Banach-limit obstruction on ℓ^∞, are treated explicitly here. The requisite background in infinite-dimensional holomorphy is in Mujica [2] and Dineen [1].

Organization. Section 2 fixes the framework and the notion of admissible holomorphicity. Section 3 proves the analytic core: tail-derivative agreement, averaging, and vanishing at constant points. Section 4 assembles the main theorem and its vector-valued corollary. Section 5 gives the counterexamples showing each hypothesis is necessary. Section 6 discusses the zero-set section problem and its open lifting lemma. Section 7 collects further remarks.

Formal verification. Every numbered mathematical statement of Sections 2–5 has been formally verified in Lean 4 against Mathlib: the definitions of Section 2; the analytic core (Lemmas 3.1 and 3.3, Proposition 3.4, Corollary 3.5, Proposition 3.6); the rigidity theorem (Theorem 4.1) together with the unconditional vanishing (Proposition 4.2); the vector-valued corollary (Corollary 4.4); both counterexamples of Section 5, including the construction of a finite-permutation-invariant Banach limit; and the exceptional-coordinate obstruction of Remark 4.3, which appears in the development as a proved theorem rather than a heuristic. The development compiles without `sorry`, and every theorem depends only on the axioms `propext`, `Classical.choice`, and `Quot.sound`. In one respect the formal development exceeds the text: Corollary 4.4(ii) is proved for normed targets with no separation hypothesis, the Hahn–Banach theorem supplying the separating functionals. The conditional Proposition 6.1 is not formalized as such; its rigidity input is Corollary 4.4, and the Lifting Lemma on which it rests is open.

The formalization was carried out with Aristotle (Harmonic) from written work orders; the audit reports are archived alongside the proofs. It repaid the effort: an earlier draft's permutation identity omitted the movement of the base point, and its local-to-global step was false outright, the development supplying an explicit counterexample; the statements above reflect those corrections.

The formalization and the audit reports are archived at Zenodo and developed on GitHub [5]:

https://doi.org/10.5281/zenodo.21944129
https://github.com/DavidVFeldman/rigidity-symmetric-holomorphic

2. Preliminaries

Let $\ell^\infty = \ell^\infty(\mathbb{N})$ be the Banach space of bounded complex sequences with the supremum norm $\|\cdot\|_\infty$. Let $c_0 \subset \ell^\infty$ be the closed subspace of sequences tending to 0, and let $c_{00} \subset c_0$ be the dense subspace of finitely supported sequences. Write e_k for the sequence with 1 in position k and 0 elsewhere, and recall $\overline{c_{00}} = c_0$ in $\|\cdot\|_\infty$.

Let S_{fin} denote the group of *finite permutations* of $\mathbb{N}$: the bijections $\sigma : \mathbb{N} \to \mathbb{N}$ fixing all but finitely many elements. It acts on $\mathbb{C}^\mathbb{N}$ by $(x \circ \sigma)_n = x_{\sigma(n)}$.

Definition 2.1. A subset $X \subset \mathbb{C}^\mathbb{N}$ is:

- *permutation-invariant* if $x \circ \sigma \in X$ for every $x \in X$ and every $\sigma \in S_{\text{fin}}$;
- *locally box-open* if for every $x \in X$ there is $r > 0$ with $x + h \in X$ for all $h \in c_{00}$ with $\|h\|_\infty < r$;
- c_{00}*-chain-connected* if for any $x, y \in X$ there is a finite sequence $x = z_0, z_1, \ldots, z_m = y$ in X with $z_{\ell+1} - z_\ell \in c_{00}$ and the segment $\{z_\ell + t(z_{\ell+1} - z_\ell) : t \in [0, 1]\}$ contained in X for each ℓ.

The distinction between c_{00}-chain-connectedness and topological connectedness is central; see Section 5. For orientation: c_0 itself is c_{00}-chain-connected (any null sequence is a sup-norm limit of finitely supported ones, but more to the point any two finitely supported sequences are joined by a single c_{00}-segment, and the general case follows by density arguments on the relevant domains), whereas ℓ^∞ is not.

Definition 2.2 (Admissible holomorphicity)**.** A function $f : X \to \mathbb{C}$ on a locally box-open, permutation-invariant set $X \subset \mathbb{C}^\mathbb{N}$ is *admissibly holomorphic* if it carries a differential datum $x \mapsto Df(x)$, where each $Df(x) : c_{00} \to \mathbb{C}$ is $\mathbb{C}$-linear, subject to:

(i) *(Coordinatewise holomorphy.)* For every $x \in X$ and every $k \in \mathbb{N}$, the map $\varepsilon \mapsto f(x+\varepsilon e_k)$ is holomorphic in a neighborhood of $0 \in \mathbb{C}$.

(ii) *(Bounded first-order expansion along c_{00}.)* For every $x \in X$ there is a constant C_x with $|Df(x)[h]| \le C_x \|h\|_\infty$ for all $h \in c_{00}$, and

$$f(x + h) = f(x) + Df(x)[h] + o(\|h\|_\infty) \qquad \text{as } h \to 0 \text{ in } c_{00}.$$

Remark 2.3 (Standard holomorphy is admissible)*.* Both standard notions of holomorphy on ℓ^∞ (or on c_0) supply Definition 2.2 with a *bounded* differential. A Fréchet-holomorphic function has a continuous Fréchet differential $Df(x)$, whose restriction to c_{00} is bounded and supplies (ii), with (i) immediate. A Gâteaux-holomorphic and locally bounded function—the standard definition in infinite dimensions, see [2, 1]— yields, via the one-variable Cauchy estimates applied coordinatewise, a bounded linear differential on finitely supported directions, giving both conditions. The boundedness in (ii) is thus automatic in every standard setting; we include it explicitly because it is exactly what the averaging step requires, and because Definition 2.2 without it does not force sup-norm continuity of f along c_{00}.

3. The Analytic Core

Throughout this section $X \subset \mathbb{C}^{\mathbb{N}}$ is locally box-open and permutation-invariant, and $f : X \to \mathbb{C}$ is admissibly holomorphic and S_{fin}-invariant.

3.1. **The permutation identity.**

Lemma 3.1 (Base-point identity). *For distinct $i, j \in \mathbb{N}$ and every $x \in X$,*

$$Df(x)[e_i] = Df(x \circ \tau_{ij})[e_j],$$

where $\tau_{ij} \in S_{\text{fin}}$ is the transposition of i and j. In particular, if $x_i = x_j$ then $x \circ \tau_{ij} = x$ and $Df(x)[e_i] = Df(x)[e_j]$.

Proof. For $\varepsilon \in \mathbb{C}$ with $|\varepsilon|$ small, local box-openness gives $x + \varepsilon e_i \in X$. A direct computation from $(y \circ \tau_{ij})_n = y_{\tau_{ij}(n)}$ shows

$$(x + \varepsilon e_i) \circ \tau_{ij} = (x \circ \tau_{ij}) + \varepsilon e_j.$$

By S_{fin}-invariance of f,

$$f(x + \varepsilon e_i) = f\big((x + \varepsilon e_i) \circ \tau_{ij}\big) = f\big((x \circ \tau_{ij}) + \varepsilon e_j\big).$$

Subtract $f(x) = f(x \circ \tau_{ij})$ (again by invariance), divide by ε, and let $\varepsilon \to 0$; coordinatewise holomorphy, condition (i), guarantees the two one-variable limits exist and equal the respective coordinate derivatives, giving $Df(x)[e_i] = Df(x \circ \tau_{ij})[e_j]$. If $x_i = x_j$ then swapping positions i, j leaves x unchanged, so $x \circ \tau_{ij} = x$ and the two base points coincide. □

Remark 3.2. The identity of Lemma 3.1 relates *different* base points whenever $x_i \neq x_j$, and one cannot conclude $Df(x)[e_i] = Df(x)[e_j]$ without further input. Equality of coordinate derivatives at a single point is available exactly along coordinates on which x is constant.

3.2. **Averaging.**

Lemma 3.3 (Cesàro vanishing). *Fix $x \in X$ and let C_x bound $Df(x)$ as in Definition 2.2(ii). For any injective sequence $i_1, i_2, \dots$ of indices,*

$$\frac{1}{N} \sum_{k=1}^{N} Df(x)[e_{i_k}] \longrightarrow 0 \qquad (N \to \infty).$$

Proof. Set $h_N = \frac{1}{N} \sum_{k=1}^{N} e_{i_k} \in c_{00}$. Distinct indices give $\|h_N\|_\infty = 1/N$. By linearity, $\frac{1}{N} \sum_{k=1}^{N} Df(x)[e_{i_k}] = Df(x)[h_N]$, and boundedness gives $|Df(x)[h_N]| \le C_x \|h_N\|_\infty = C_x/N \to 0$. □

3.3. **Vanishing at eventually-constant points.**

Say $x \in \mathbb{C}^{\mathbb{N}}$ is *eventually constant with value* c if $x_n = c$ for all but finitely many n; write $\operatorname{supp}_c(x) = \{n : x_n \neq c\}$, a finite set, for its *exceptional set.*

Proposition 3.4 (Tail vanishing). *Let $x \in X$ be eventually constant with value c. Then $Df(x)[e_i] = 0$ for every index $i \notin \operatorname{supp}_c(x)$; that is, the coordinate derivative vanishes along the constant tail.*

Proof. Let $T = \mathbb{N} \setminus \operatorname{supp}_c(x)$ be the (cofinite) tail, so $x_i = c$ for all $i \in T$. For $i, j \in T$ we have $x_i = x_j = c$, so Lemma 3.1 gives $Df(x)[e_i] = Df(x)[e_j]$. Hence there is a scalar λ with $Df(x)[e_i] = \lambda$ for all $i \in T$. Since T is infinite, choose an injective sequence $i_1, i_2, \dots$ inside T; each term of the Cesàro average in Lemma 3.3 equals λ, so the average is constantly λ while tending to 0. Therefore $\lambda = 0$, i.e. $Df(x)[e_i] = 0$ for every $i \in T$. □

Corollary 3.5 (Vanishing at constant points)**.** *If the constant sequence $x \equiv c$ lies in X, then $Df(x)[h] = 0$ for all $h \in c_{00}$.*

Proof. A constant sequence has empty exceptional set, so Proposition 3.4 gives $Df(x)[e_k] = 0$ for every k. Any $h \in c_{00}$ is a finite combination $h = \sum_k h_k e_k$, and linearity of $Df(x)$ gives $Df(x)[h] = 0$. □

3.4. **Constancy along finitely supported segments.**

Proposition 3.6 (Segment rigidity)**.** *Let $x \in X$ and $h \in c_{00}$ be such that the segment $\gamma(t) = x + t\,h$ lies in X for all $t \in [0,1]$, and suppose $Df(\gamma(t))[h] = 0$ for every $t \in [0,1]$. Then $f(x+h) = f(x)$.*

Proof. Consider $\varphi(t) = f(x + t\,h)$ on $[0,1]$. Fix $t_0 \in [0,1]$. For small real s, the first-order expansion at the base point $\gamma(t_0)$, applied to the finitely supported increment $s\,h$, gives

$$\varphi(t_0+s) - \varphi(t_0) = f\big(\gamma(t_0) + s\,h\big) - f(\gamma(t_0)) = s\,Df(\gamma(t_0))[h] + o(|s|\,\|h\|_\infty) = o(|s|),$$

using $Df(\gamma(t_0))[h] = 0$. Hence φ is differentiable at t_0 with $\varphi'(t_0) = 0$. As $t_0 \in [0,1]$ was arbitrary, φ has identically zero derivative on $[0,1]$ and is therefore constant; in particular $\varphi(1) = \varphi(0)$, i.e. $f(x+h) = f(x)$. □

4. The Rigidity Theorem

The natural domains for the main theorem are those sitting inside a single translate of c_0—equivalently, those whose points share one eventual value. On such a domain the c_{00}-directional derivative structure controls the function, and the theorem below gives constancy once the differential vanishes on c_{00} throughout. Proposition 4.2 then records the extent to which that hypothesis holds without further assumption.

Theorem 4.1 (Rigidity on null-sequence domains)**.** *Fix $c \in \mathbb{C}$ and let $X \subset (c \cdot \mathbf{1}) + c_0$ be nonempty, locally box-open, permutation-invariant, and c_{00}-chain-connected, where $\mathbf{1} = (1,1,\dots)$. Let $f : X \to \mathbb{C}$ be admissibly holomorphic with bounded differential and invariant under all finite permutations. Suppose moreover that $Df(x)[h] = 0$ for all $x \in X$ and all $h \in c_{00}$. Then f is constant on X.*

Proof. Given $x, y \in X$, take a c_{00}-chain $x = z_0, \dots, z_m = y$ as in Definition 2.1, with each increment $h_\ell := z_{\ell+1} - z_\ell \in c_{00}$ and each segment inside X. Along the ℓ-th segment the standing hypothesis gives $Df(z_\ell + t\,h_\ell)[h_\ell] = 0$ for all $t \in [0,1]$, so Proposition 3.6 yields $f(z_{\ell+1}) = f(z_\ell)$. Chaining the equalities gives $f(y) = f(x)$. As x, y were arbitrary, f is constant. □

The theorem carries the hypothesis $Df|_{c_{00}} \equiv 0$ because it is not implied, at every point of an eventually-constant domain, by the remaining hypotheses. What symmetry alone yields is the value of the differential at constant points and along constant tails.

Proposition 4.2 (Vanishing at constant points and tails)**.** *In the setting of Section 3, $Df(c \cdot \mathbf{1})[h] = 0$ for every $h \in c_{00}$ whenever $c \cdot \mathbf{1} \in X$* (Corollary 3.5)*; and for any eventually-constant $x \in X$ with value c, $Df(x)[e_i] = 0$ for every tail index i with $x_i = c$* (Proposition 3.4)*.*

Proposition 4.2 marks how far the hypothesis $Df|_{c_{00}} \equiv 0$ of Theorem 4.1 holds without further assumption. In particular, on a domain of constant sequences every coordinate is a tail coordinate, so the hypothesis holds throughout and f is constant unconditionally. At the exceptional coordinates of a non-constant point the differential is *not* controlled by this argument, for a definite reason.

Remark 4.3 (The exceptional-coordinate obstruction). Removing an exceptional coordinate k of a point x (where $x_k \neq c$) by passing to the "zeroed" point p obtained from x by resetting the k-th entry to c—at which k is a tail coordinate—does not transport the vanishing back to x. The base-point identity Lemma 3.1 gives $Df(x)[e_k] = Df(x \circ \tau_{kj})[e_j]$, but $x \circ \tau_{kj}$ merely *moves* the exceptional value from position k to position j; it is not a point at which k is a tail coordinate. Concretely, $x \circ \tau_{kj}$ and $p \circ \tau_{kj}$ disagree at position j (values x_k versus c), so the identity at x does not relate to p. Since each eventually-constant sequence has only finitely many exceptional coordinates, the averaging argument of Lemma 3.3—which requires infinitely many equal coordinate derivatives at a *single* base point—does not apply to the exceptional directions. Whether $Df|_{c_{00}} \equiv 0$ holds at every point under a stronger analytic hypothesis—for instance continuity of $x \mapsto Df(x)$ in the supremum norm, available for Fréchet-holomorphic f—is open.

4.1. **The vector-valued corollary.**

Corollary 4.4. *Let X be as in Theorem 4.1, and let Y be a locally convex complex topological vector space. Suppose $\Sigma : X \to Y$ is S_{fin}-equivariant, $\Sigma(x \circ \sigma) = \Sigma(x)$, and such that for every $\lambda \in Y^*$ the scalar function $\lambda \circ \Sigma$ satisfies the hypotheses of Theorem 4.1. Then:*

(i) *$\lambda \circ \Sigma$ is constant for every $\lambda \in Y^*$;*
(ii) *if Y^* separates the points of Y, then Σ is constant.*

Proof. Each $\lambda \circ \Sigma$ is S_{fin}-invariant because Σ is equivariant, so Theorem 4.1 gives (i). For (ii), if $\Sigma(x) \neq \Sigma(y)$ some $\lambda \in Y^*$ separates the values, contradicting (i). □

5. Sharpness: the Hypotheses Are Necessary

The hypothesis confining the domain to a single c_0-coset, which is to say the hypothesis of c_{00}-chain-connectedness, cannot be dropped, as the following two examples show. Both are genuinely symmetric and genuinely holomorphic; what fails is precisely the passage from c_{00}-local to global constancy.

Example 5.1 (Two cosets: a disconnected witness). For $c \in \{0, 1\}$ let $S_c = \{x \in \ell^\infty : x_n = c \text{ for all but finitely many } n\}$ be the eventually-c sequences, and put $X = S_0 \cup S_1$. Each S_c is permutation-invariant and locally box-open, and a c_{00}-perturbation of a point of S_c remains in S_c; thus X is locally box-open and permutation-invariant. Define $f : X \to \mathbb{C}$ by $f \equiv 0$ on S_0 and $f \equiv 1$ on S_1. Then f is S_{fin}-invariant, and it is admissibly holomorphic with $Df \equiv 0$ (it is locally constant along c_{00}, since c_{00}-perturbations do not change the eventual value). Yet f is not constant. The obstruction is exactly that S_0 and S_1 are distinct cosets of c_0: no finite chain of c_{00}-steps joins them, so X is not c_{00}-chain-connected.

Example 5.2 (A connected witness: Banach limits). Let $L \in (\ell^\infty)^*$ be a Banach limit: a continuous linear functional with $L(\mathbf{1}) = 1$, invariant under the shift, and hence under every finite permutation, and extending the ordinary limit on convergent sequences. Being continuous and linear, L is entire on ℓ^∞, so it is admissibly holomorphic with $DL(x) = L$ for all x; its differential is bounded (by $\|L\|$). Since L vanishes on $c_0 \supset c_{00}$, we have $DL(x)|_{c_{00}} = 0$ at every point—exactly as the analytic core predicts. Nonetheless L is *not* constant: $L(0) = 0$ while $L(\mathbf{1}) = 1$.

Here the domain is all of ℓ^∞, which *is* connected (indeed convex), so mere topological connectedness does not suffice for rigidity. What fails is c_{00}-chain-connectedness: 0 and $\mathbf{1}$ are at sup-distance 1 from one another across the quotient ℓ^∞ / c_0, and no finite c_{00}-chain bridges them. This example shows that the restriction to null-sequence domains in Theorem 4.1 is not a technical convenience but a necessity.

Remark 5.3 (Sharpness of the topology). A complementary degeneration occurs if the supremum norm is replaced by a summing norm. On $\ell^1 \cap D^{\mathbb{N}}$ the averages $h_N = \frac{1}{N}\sum_{k=1}^{N} e_{i_k}$ have $\|h_N\|_1 = 1$ for all N, so the Cesàro step (Lemma 3.3) fails, and indeed $f(x) = \sum_k x_k$ is a non-constant symmetric holomorphic function. Thus a topology in which averaged coordinate perturbations become small—such as the supremum norm on c_0—is essential to the averaging mechanism.

6. Application: the Zero-Set Section Problem

We indicate how the rigidity theorem bears on a natural geometric question, and we state precisely the point at which the argument becomes conditional.

Let $\mathcal{H} = \mathcal{O}(B(0,1))$ be the Fréchet space of holomorphic functions on the open unit disk with the compact-open topology, and let $\mathcal{D}$ be the space of admissible infinite discrete subsets of $B(0,1)$—sequences with no limit point inside the disk. By the Weierstrass factorization theorem the zero-set map

$$\pi : \mathcal{H} \setminus \{0\} \longrightarrow \mathcal{D}, \qquad f \longmapsto (\text{zero multiset of } f),$$

is surjective. One asks whether π admits a holomorphic cross-section $\sigma : \mathcal{D} \to \mathcal{H}$ with $\pi \circ \sigma = \mathrm{id}$.

Model $\mathcal{D}$ as a quotient Seq/S_∞, where Seq is the space of locally finite sequences in $B(0,1)$; the naturally arising domain of sequences is c_{00}-chain-connected, since admissible sets are altered one point at a time by finitely supported moves. The rigidity theorem then yields a conditional obstruction.

Proposition 6.1 (Conditional obstruction)**.** *Suppose a cross-section $\sigma : \mathcal{D} \to \mathcal{H}$ lifts to an S_{fin}-equivariant map $\widetilde{\sigma} : X \to \mathcal{H}$ on the associated c_{00}-chain-connected sequence domain X, with $\lambda \circ \widetilde{\sigma}$ admissibly holomorphic (bounded differential, differential vanishing on c_{00}) for every evaluation functional $\lambda \in \mathcal{H}^*$. Then no such σ exists.*

Proof. Since $\mathcal{H}$ separates points via evaluations, Corollary 4.4(ii) forces $\widetilde{\sigma}$ to be constant, say $\widetilde{\sigma} \equiv f_0$. But then $\pi(f_0) = \pi(\widetilde{\sigma}(x))$ would equal the varying zero set of x for every $x \in X$, which is absurd. $\square$

The hypothesis is a genuine lifting condition, stated precisely below.

> **Open Problem (Lifting Lemma).** Let Y be a locally convex space. Under a suitable complex structure on $\mathcal{D} = \mathrm{Seq}/S_\infty$, does every holomorphic map $F : \mathcal{D} \to Y$ lift to an S_{fin}-equivariant map $\widetilde{F} : \mathrm{Seq} \to Y$ whose scalarizations are admissibly holomorphic with differentials vanishing on c_{00}?

An affirmative answer would, through Corollary 4.4, upgrade Proposition 6.1 to the unconditional statement that π admits no holomorphic section on all of $\mathcal{D}$. We leave this as the natural next target.

Remark 6.2 (A complementary unconditional obstruction). Independently of the lifting problem, there is no section of the special *product* form $\sigma(A) = \prod_{a \in A} E(z,a)$ for a fixed holomorphic factor E: symmetry forces the factor to be uniform, a uniform factor imposes one fixed summability condition on the zero set for convergence, and $\mathcal{D}$ contains admissible sets violating any such condition. Thus symmetric product sections fail for convergence reasons unconditionally, while any symmetric non-product section would be obstructed by rigidity, conditionally on the lifting.

7. Further Remarks

(1) *What symmetry alone gives.* The analytic core (Section 3) is valid on any locally box-open permutation-invariant domain, without chain-connectedness: it yields tail-derivative vanishing at every point and full vanishing at constant points. Chain-connectedness enters only to globalize, and Example 5.2 shows it cannot be omitted.

(2) *Relation to de Finetti's theorem.* The result is an analytic cousin of de Finetti's theorem: exchangeability of a measure on $\{0,1\}^{\mathbb{N}}$ forces dependence only on "collective" statistics, whereas here symmetry forces a holomorphic function on a null-sequence domain to depend on nothing at all. The Banach-limit example marks the boundary: on ℓ^∞, symmetric "limit-like" functionals persist, just as tail σ-fields carry exchangeable information the mixture representation cannot see.

(3) *Comparison with finite dimensions.* For each finite n the S_n-invariant holomorphic functions on $\mathbb{C}^n$ form the polynomial algebra on $e_1, \dots, e_n$, each a finite and hence entire sum. In infinite dimensions these sums diverge on generic sequences; Proposition 3.4 is the precise derivative-level expression of that divergence along the constant tail.

(4) *Positive results on smaller domains.* On $\ell^2 \cap D^{\mathbb{N}}$ the power sums $p_k = \sum_n a_n^k$ converge and are holomorphic for $k \geq 3$, providing non-constant symmetric holomorphic functions. Rigidity is a feature of the c_0 geometry with its averaging mechanism, not of symmetry as such.

(5) *A topological analogue.* The averaging argument is not specific to holomorphy. The same reasoning shows that a permutation-invariant function on a c_{00}-chain-connected domain in c_0 that is Lipschitz and Gâteaux-differentiable along c_{00}, with bounded differential, is constant. Holomorphy is used only to produce the coordinate derivatives in Lemma 3.1; the heart of the matter is the metric fact $\|h_N\|_\infty = 1/N \to 0$ together with the coset structure of c_0 inside ℓ^∞.

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